LATTICE BASIS REDUCTION

2. GAUSS'S ALGORITHM

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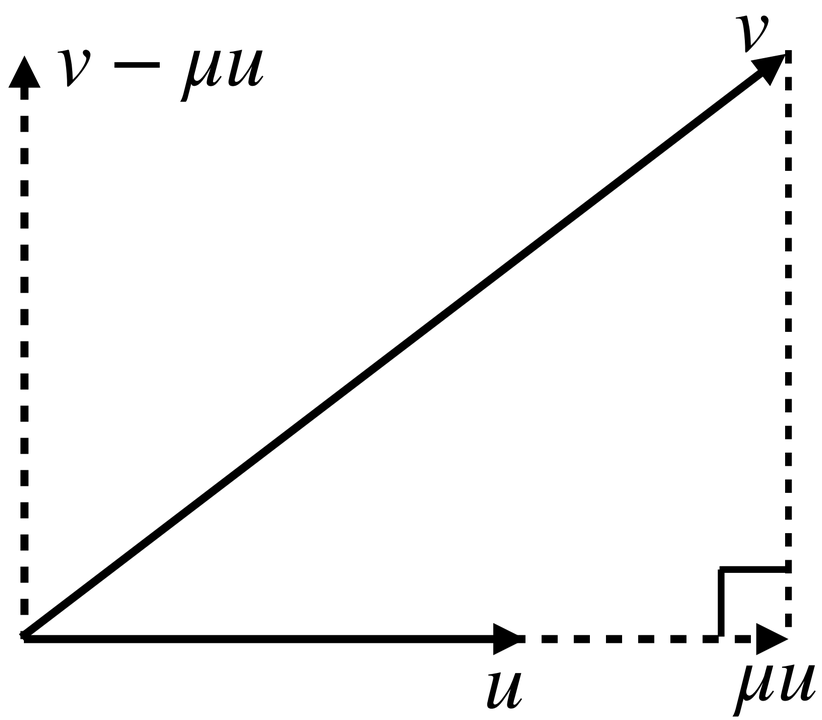
Problem statement

- **Recall**:
  - $\|v\| = \sqrt{v_1^2 + \cdots + v_n^2}$ denotes the Euclidean length of a vector $v = (v_1, \ldots, v_n) \in \mathbb{R}^n$.
  - First successive minimum: $\lambda_1(L)$ denotes the length of a shortest nonzero vector in a lattice L.
  - Second successive minimum: $\lambda_2(L)$ is the smallest real number r for which there exists two linearly independent vectors in L, each of which has length $\leq r$.
- **Input**: A basis $[u, v]$ for a two-dimensional integer lattice $L \subseteq \mathbb{Z}^2$, where $\|u\| \leq \|v\|$.
- **Output**: A basis $[u', v']$ for L with $\|u'\| = \lambda_1(L)$ and $\|v'\| = \lambda_2(L)$.
  - **Note**: $[u', v']$ is a smallest basis of L.

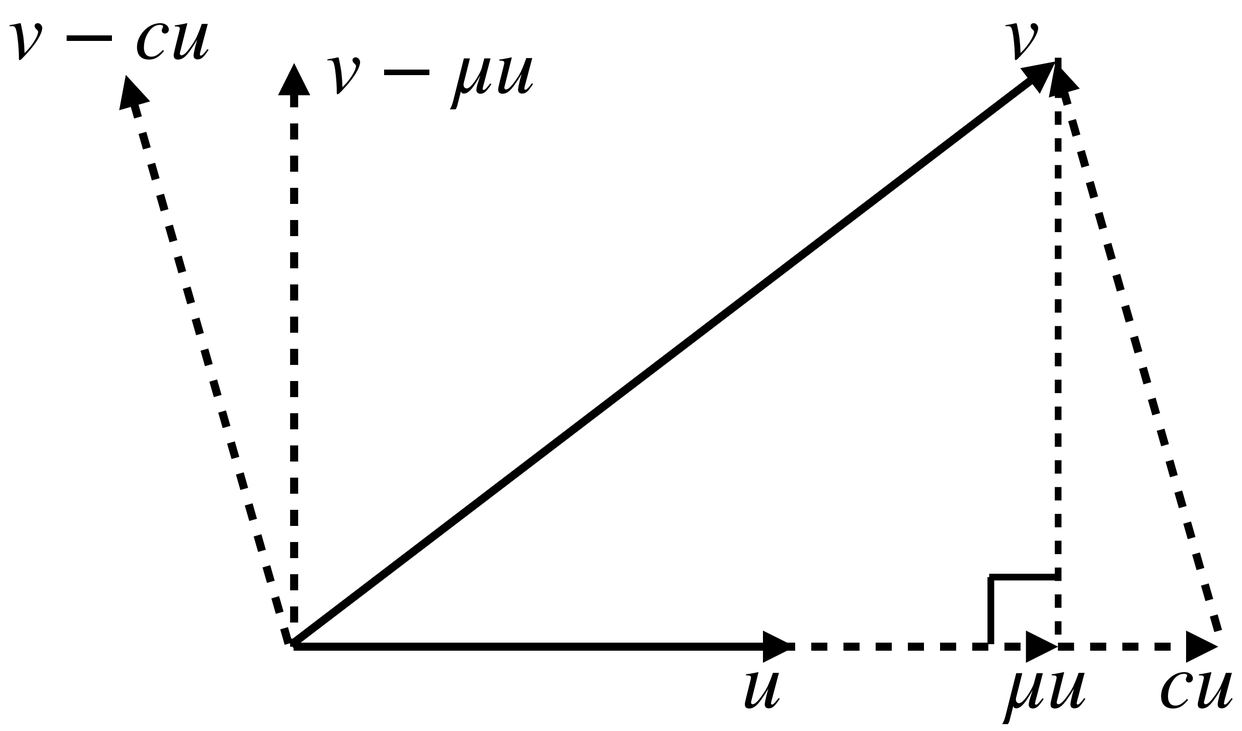
Projecting a vector onto another vector

- Let $u, v \in \mathbb{R}^2 \backslash \{0\}$, and suppose that v is not a multiple of u.
- We wish to determine the projection of v onto $\langle u \rangle$ and $\langle u \rangle^\perp$.
- Let $w \in \mathbb{R}^2$ with $w \neq 0$ and $\langle u, w \rangle = 0$.
    - The dot product (or inner product) of $u = (u_1, \ldots, u_n)$, $w = (w_1, \ldots, w_n) \in \mathbb{R}^n$ is $\langle u, w \rangle = u_1 w_1 + \cdots + u_n w_n$.
- Then $[u, w]$ is an orthogonal basis for $\mathbb{R}^2$.
- Now, write $v = \mu u + \omega w$, where $\mu, \omega \in \mathbb{R}$.
- Then $\langle u, v \rangle = \langle u, \mu u + \omega w \rangle = \mu \langle u, u \rangle + \omega \langle u, w \rangle = \mu \|u\|^2$, so $\mu = \langle u, v \rangle / \|u\|^2$. Also, $\langle u, v - \mu u \rangle = 0$.
- Thus $\Pi_{\langle u \rangle}(v) = \mu u$ where $\mu = \langle u, v \rangle / \|u\|^2$, and $\Pi_{\langle u \rangle^\perp}(v) = v - \mu u$.



Gauss's algorithm: main idea

- Gauss described his algorithm in 1801.
- Actually, the algorithm was first discovered by Lagrange in 1773
- **Input**: A basis $[u, v]$ for a two-dimensional integer lattice $L \subseteq \mathbb{Z}^2$, where $\|u\| \leq \|v\|$.



- $\mu = \langle u, v \rangle / \|u\|^2$.
- Note: μ is not necessarily an integer; if it isn't, then $v - \mu u \notin L$
- Let $c = \lfloor \mu \rceil$ be the closest integer to μ, with ties broken by rounding towards 0 (so $\lfloor 3.5 \rceil = 3$ and $\lfloor -3.5 \rceil = -3$).
- Then $v - cu \in L$, and $[u, v - cu]$ is a shorter basis for L.

Gauss's algorithm

- **Input**: A basis $[u, v]$ for a lattice $L \subseteq \mathbb{Z}^2$, where $\|u\| \leq \|v\|$.
- **Output**: A basis $[u', v']$ for L with $\|u'\| = \lambda_1(L)$ and $\|v'\| = \lambda_2(L)$.

1. $c \leftarrow 1$.
2. While $c \neq 0$ do
   (a) Compute $\mu = \langle u, v \rangle / \|u\|^2$ and $c = \lfloor \mu \rceil$.
   (b) Set $v \leftarrow v - cu$.
   (c) If $\|u\| > \|v\|$ then
      - Swap u and v.
3. Return($[u, v]$).

Running time

- **Claim 1**. The output of Gauss's algorithm is a basis of L.
- **Claim 2**. The algorithm terminates.
- **Claim 3**. The number of iterations is $O(\log X)$, where $X = \|v\|$.
- **Claim 4**. All integers in the algorithm are $\leq X$, and consequently the running time of Gauss's algorithm is $O(\log^3 X)$ bit operations.
    - Nguyen and Stehlé improved the running time bound to $O(\log^2 X)$ bit operations.

Example 1

- Consider the lattice L with basis vectors $u = (3,5)$, $v = (4,7)$.

#	u	$\|u\|^2$	v	$\|v\|^2$	μ	c	$v - cu$
1	(3,5)	34	(4,7)	65	47/34	1	(1,2)
2	(1,2)	5	(3,5)	34	13/5	3	(0,-1)
3	(0,-1)	1	(1,2)	5	-2	-2	(1,0)
4	(0,-1)	1	(1,0)	1	0	0	(1,0)

- A short basis for L is $[u', v']$ where $u' = (0,-1)$ and $v' = (1,0)$.

Example 2

- Consider the lattice L with basis vectors $u = (47928,63649)$, $v = (68827,91412)$.

#	u	$\|u\|^2$	v	$\|v\|^2$	μ	c	$v - cu$
1	(47928, 63649)	6348288385	(68827, 91412)	13093309673	$\frac{9117022844}{6348288385}$	1	(20899, 27763)
2	(20899, 27763)	1207552370	(47928, 63649)	6348288385	$\frac{2768734459}{1207552370}$	2	(6130, 8123)
3	(6130, 8123)	103560029	(20899, 27763)	1207552370	$\frac{353629719}{103560029}$	3	(2509, 3394)
4	(2509, 3394)	17814317	(6130, 8123)	103560029	$\frac{42949632}{17814317}$	2	(1112, 1335)
5	(1112, 1335)	3018769	(2509, 3394)	17814317	$\frac{7320998}{3018769}$	2	(285, 724)
6	(285, 724)	605401	(1112, 1335)	3018769	$\frac{1283460}{605401}$	2	(542, -113)
7	(542, -113)	306533	(285, 724)	605401	$\frac{72658}{306533}$	0	(285, 724)

- A short basis for L is $[u', v']$ where $u' = (542,-113)$ and $v' = (285,724)$.

First successive minimum

- **Claim 5**. Let $[u, v]$ be the output of Gauss's algorithm. Then $\|u\| = \lambda_1(L)$.
- **Proof**. Let $w \in L \backslash \{0\}$, and write $w = au + bv$ where $a, b \in \mathbb{Z}$ (not both 0). We have

$$\begin{aligned}
\|w\|^2 = \|au + bv\|^2 &= \langle au + bv, au + bv \rangle \\
&= a^2\|u\|^2 + 2ab\,\langle u, v\rangle + b^2\|v\|^2 \\
&\geq a^2\|u\|^2 - 2\,|ab|\,\langle u, v\rangle + b^2\|u\|^2 \quad (\text{since } \|v\| \geq \|u\| \text{ and } ab \geq -|ab|) \\
&\geq a^2\|u\|^2 - |ab| \cdot \|u\|^2 + b^2\|u\|^2 \quad (\text{since } |\mu| = |\langle u, v\rangle|/\|u\|^2 \leq \frac{1}{2}) \\
&= (a^2 - |a| \cdot |b| + b^2)\|u\|^2 \\
&\geq \|u\|^2.
\end{aligned}$$

Thus, $\|w\| \geq \|u\|$. $\square$

Let $a, b \in \mathbb{R}$. Then $f(a, b) = a^2 - ab + b^2 = \left(a - \frac{b}{2}\right)^2 + \frac{3}{4}b^2 \geq 0$, with $f(a, b) = 0$ iff $a = b = 0$.
Thus, if $a, b \in \mathbb{Z}$ (not both 0) then $a^2 - |a| \cdot |b| + b^2 \geq 1$.

Second successive minimum

- **Claim 6**. Let $[u, v]$ be the output of Gauss's algorithm. Then $\|v\| = \lambda_2(L)$.

- **Proof**. Let $w \in L$ and suppose that w is not a multiple of u.
Then we can write $w = au + bv$ where $a, b \in \mathbb{Z}$ with $b \neq 0$. We have

$$\begin{aligned}
\|w\|^2 = \|au + bv\|^2 &= \langle au + bv, au + bv \rangle = a^2\|u\|^2 + 2ab\langle u, v\rangle + b^2\|v\|^2 \\
&= b^2(\|v\|^2 - \|u\|^2) + (a^2 + b^2)\|u\|^2 + 2ab\langle u, v\rangle \\
&\geq b^2(\|v\|^2 - \|u\|^2) + (a^2 - ab + b^2)\|u\|^2 \quad (\text{since } |\mu| = |\langle u, v\rangle|/\|u\|^2 \leq \tfrac{1}{2}) \\
&\geq b^2(\|v\|^2 - \|u\|^2) + \|u\|^2 + (\|v\|^2 - \|v\|^2) \quad (\text{since } a^2 - ab + b^2 \geq 1) \\
&= (b^2 - 1)(\|v\|^2 - \|u\|^2) + \|v\|^2 \\
&\geq \|v\|^2 \quad \text{since } b \neq 0 \text{ and } \|v\| \geq \|u\|, \quad \text{so } \|w\| \geq \|v\|.
\end{aligned}$$

Finally, suppose that $[u', v']$ are linearly independent lattice vectors. Then u', v' can't both be multiples of u. Suppose that v' is not a multiple of u; then $\|v'\| \geq \|v\|$ by what we have shown above. If u' is a multiple of u, then $\|u'\| \geq \|u\|$; otherwise $\|u'\| \geq \|v\|$. The result follows. $\square$

Higher-dimension lattices

- **Definition**. A basis $[u, v]$ for a lattice $L \subseteq \mathbb{Z}^2$ is reduced if
$$\|u\| \leq \|v\| \text{ and } |\mu| \leq \frac{1}{2} \text{ where } \mu = \langle u, v \rangle / \|u\|^2.$$
- Gauss's algorithm produces a reduced basis for a lattice $L \subseteq \mathbb{Z}^2$.
  - As we've seen, this basis is a shortest possible basis for L.
- This notion of a reduced basis extends naturally to lattices in $\mathbb{Z}^n$ where $n \geq 3$.
  - However, a lattice basis that is reduced is not necessarily a short basis.
- Lovász introduced a stronger notion of reduced lattice basis.
  - The vectors in such a basis are relatively short.
  - The LLL algorithm efficiently finds such a basis.